

6. Project Testing Phase

- The Project Testing Phase verifies whether the spreadsheet data-import process works correctly and meets the defined requirements. Testing is essential because successful configuration does not automatically guarantee that all records will be imported correctly.
 - The first testing activity is field mapping testing. Each source field should be checked against its corresponding target field. For example, the Employee Name from the spreadsheet should appear in the Name field of the target table. The same verification should be performed for email, department, location, employee ID, and other fields.

The second activity is data accuracy testing. A small number of spreadsheet records should be compared with the records created in the target table. The values should match unless an intentional transformation has been applied.
 - The third activity is duplicate testing. Duplicate data should be introduced in a controlled test environment to verify how the Transform Map handles existing records. If a coalesce field is configured, the expected update behavior should be verified.
 - The fourth activity is invalid-data testing. Examples of invalid data include missing mandatory fields, incorrect email formats, invalid dates, or unknown reference values. The system should handle these records according to the defined requirements.
- Another important test is large-data testing. After the Transform Map works correctly with a small sample, a larger spreadsheet should be imported. This helps identify performance problems and processing issues.

Testing can be documented using a test-case table:

Test Case

Expected Result

Status

Valid spreadsheet record

Record imported successfully

Pass

Missing mandatory field

Record handled according to validation rule

Pass

Existing Employee ID

Existing record updated

Pass

New Employee ID

New record created

Pass

Invalid email

Invalid data identified

Pass

Multiple records

All valid records processed

Pass

After testing, any errors should be analyzed and corrected. The Transform Map may need to be modified if field mappings, transformation logic, or validation rules are incorrect.

Testing should continue until the import process produces consistent and expected results. The final test results should be included in the project documentation as evidence that the solution has been verified.